	The total number of marks available for this examination paper is 80. It contributes 20% to the Advanced GCE in Physics.
Calculator	It is recommended that students have access to a scientific calculator for this paper.
Formulae sheet	Students will be provided with the formulae sheet shown in *Appendix 8*: *Formulae*. Any other physics formulae that are required will be stated in the question paper.

10.3 Transport on Track (TRA)

A study of a modern rail transport system with an emphasis on safety and control:

- track circuits and signalling
- sensing speed
- mechanical braking
- regenerative and eddy current braking
- crash-proofing.

Students use mathematical models to describe the behaviour of moving vehicles and to model electromagnetic induction and capacitor discharge.

There are opportunities to develop information and communication technology skills.

There are opportunities for students to discuss ethical, environmental and other issues relating to decisions about transport taken by government, by transport companies and by individuals.

Students will be assessed on their ability to:	**Suggested experiments**
73 use the expression $p = mv$	
74 investigate and apply the principle of conservation of linear momentum to problems in one dimension	Use of, for example, light gates and air track to investigate momentum
75 investigate and relate net force to rate of change of momentum in situations where mass is constant (Newton's second law of motion)	Use of, for example, light gates and air track to investigate change in momentum
78 explain and apply the principle of conservation of energy, and determine whether a collision is elastic or inelastic	
87 investigate and use the expression $C = Q/V$	